MILITARY SPECIFICATION

SWITCH, ACCELEROMETER, ATOMIC WEAPON: XM49

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of Inertia actuated integrating electromechanical switch device entitled Acceleration Switch (MIAD) which is used with the Honest John warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure for.

DRAWINGS

ORDANANCE CORPS

8839204	- Frame.
8839205	- Weight Timing.
8839206	- Weight Setback.
8839210	- Shaft.
8839211	- Actuator Switch Monitor.
8839212	- Switch Firing Actuator (Left).
8839213	- Switch Firing Actuator (Right).
8839234	- Sleeve.
8839236	- Delay Assembly (Modified).

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8857851 - Switch, Accelerometer, Atomic Weapon: XM49.
ET-8857851 - Equipment Tabulation.
8857967 - Spring, Bias.
8857968 - Frame Assembly.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The switch shall comply with all requirements specified on the drawing (dwg) 8857851, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. The contractor shall assign serial numbers as required by drawings. The contractor may select any number which complies with the drawing requirement. These numbers shall not be duplicated.

3.5 Circuit resistance.

3.5.1 Safe Condition. The resistance between terminals 2 and 5 shall not exceed 0.035 ohm when applying a current of 100 milliamperes (ma) maximum (max). This test shall be conducted as specified in 4.4.1.

3.5.2 Operated condition. The resistance between terminals 1 and 3 and between terminals 4 and 6 shall not exceed 0.030 ohms when tested as specified in 4.4.1.

3.6 Insulation resistance.

3.6.1 Safe condition. The insulation resistance between each terminal and all other terminals except terminal 2 and between all terminals and the case shall be not less than 100 meghoms (mego) when tested as 500 plus or minus (±) 50 volts direct current (vdc) as specified in 4.4.2.

3.6.2 Operated condition. The insulation resistance between each terminal and all other terminals except 1 and 4 and between each terminal and the case shall be not less than 100 mego when tested at 500 ± 50 vdc as specified in 4.4.2.

3.7 Current carrying capability. With the switch in the operated condition, the contacts, terminals 1-3 and 4-6 in parallel, shall be capable of carrying the load current specified on Figure 2. The maximum voltage drop between terminals 1 and 3, and between terminals 4 and 6 shall not exceed 2.0 volts during the 22 second load cycle. Testing shall be conducted as specified in 4.4.3.

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3.8 Reset capability. The switch shall be capable of being reset form the operated to safe condition by pressing the flexible diaphram as specified in 4.4.4.

3.9 Environmental testing (see 4.4.5).

3.9.1 Acceleration @ 80°F. The switch and acceleration field shall be oriented in the operation direction as shown in Figure 1.

3.9.1.1 Safe condition. The switch shall be subjected to a force of 6.0 + 0.5 gravity units (g) applied for eight seconds minimum (min). Electrical continuity between terminals 2 and 5 shall not be broken when tested as specified in 4.4.5.1.1.

3.9.1.2 No load operation. The switch shall be subjected to acceleration levels in Tables I or II as appropriate. The switch operation shall establish electrical continuity between terminals 1 and 3, and between terminals 4 and 6, within the times specified at the specifc acceleration levels in Tables I or II. The circuit between terminals 2 and 5 shall open within the times specified at the specific acceleration levels in Tables I or II. The switch shall remain in the operated condition until manually reset. The switch shall meet the requirements of 3.5.2 and 3.6.2 after the last acceleration test in either Table I or II. The test shall be conducted as specified in 4.4.5.1.2. (NOTE: Switches calibrated to meet the requirements of either table shall not be required to meet the requirements of the other table).

Table I (Part No. 8857851-1)

Test	Acceleration Level (g) - 1 run ea level	Time to Operate (seconds) Terminals (1 - 3) (4 - 6) min.	Terminals (1 - 3) (4 - 6) max.	Monitor Opening Time (Seconds) Terminals (2 - 5) max.
1	15.0 ± 0.4	-	3.00	0.23
2	25.0 ± 0.6	1.40	1.72	0.17
3	40.0 ± 0.8	1.09	1.33	0.13
4	50.0 ± 0.9	0.96	1.18	0.12
5	60.0 ± 1.2	0.86	1.07	0.11

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Table II (Part No. 8857851-2)

Test	Acceleration Level (g) - 1 run ea level	Time to Operate (Seconds) Terminals (1 - 3) (4 - 6) min.	Terminals (1 - 3) (4 - 6) max.	Monitor Opening Time (Seconds) Terminals (2 - 5) max.
1	13.0 ± 0.3	-	3.25	0.32
2	15.0 ± 0.4	2.11	2.53	0.25
3	38.0 ± 0.8	1.30	1.57	0.16

3.9.1.3 Current load operation. The switch shall withstand a force of 25 ± 0.6 g for part No. 88578151-1 and 15.0 ± 0.4 g for part No. 8857851-2. The switch contacts, terminals 1-3 and 4-6, in parallel, shall close and then carry a current of 20 ± 1 amperes resistive dc load for 60 ± 40 milliseconds. Immediately following this pulse, the current shall be reduced to 5.5 ± 0.3 ampere resistive dc load for a period of 22 seconds min. The parallel resistance shall not exceed 0.015 ohm subsequent to a 0.010 second period after closure of the first contact. Testing shall be as specified in 4.4.5.1.3.

3.9.2 Operating requirements at - 40°F.

3.9.2.1 Safe condition. The switch shall meet the requirements of 3.9.1.1 at -40 ± 5°F degrees Fahrenheit (°F). Testing shall be conducted per 4.4.5.2.1.

3.9.2.2 No load operation. The switch shall meet the requirements of 3.9.1.2 when subjected to test #2 of Tables I or II as applicable and tested at -40 ± 5°F. Testing shall be conducted as specified in 4.4.5.2.2.

3.9.3 Operating Requirements at 160 °F.

3.9.3.1 Safe condition. The switch shall meet the requirements of 3.9.1.1 at 160 ± 5°F. Testing shall be conducted as specified in 4.4.5.3.1.

3.9.3.2 No load operation. The switch shall meet the requirements of 3.9.1.2 applicable to test #2 in Table I or II, as appropriate, at 160 ± 5°F. Testing shall be conducted per 4.4.5.3.2.

3.9.4 Sealing.

3.9.4.1 Test No. 1. The switch shall not leak gas from within at a rate greater than 1 x 10^{-7} cubic centimeter per second when tested as specified in 4.4.5.4.

3.9.4.2 Test No. 2. The temperature of the switch shall be raised to 80 ± 20°F. It shall be immersed for 30 seconds min. in water heated to 175 ± 15°F. Evidence of bubbles shall indicate leakage. Testing shall be as specified in 4.4.5.4.

3.9.5 Impact shock.

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3.9.5.1 Safe condition. The switch shall withstand 40 impact shocks in the operating direction (see Figure 1) of 22 ± 2 g peak amplitude for 10 ± 2 milliseconds (ms) at the 10 g level. Following the shock test, and prior to removing the shock producing device, the parallel circuits between terminals 1 and 3 and between terminals 4 and 6 shall show no evidence of closure. The circuit between terminals 2 and 5 shall indicate closure. Testing shall be as specified in 4.4.5.5.1.

3.9.5.2 Operated condition. The switch shall withstand three impact shocks, in the direction opposite to that required for operation (see Fig. 1) at 80 ± 8 g peak amplitude for 3 ± 1 ms at base amplitude. During the 5 ms internal, immediately following the initiation of the shock pulse, continuity shall exist between terminals 1 and 3, and between 4 and 6. Testing shall be as specified in 4.4.5.5.2.

3.9.6 Vibration.

3.9.6.1 Safe condition. The switch shall withstand transportation vibration at the frequencies, amplitudes, and for the time specified in Table III without evidence of damage or deterioration affecting function. Testing shall be as specified in 4.4.5.6.1.

Table III

Frequency range (Hertz)	Double displacement or peak g level	Cycle Time to cover freq. range-(minutes)	Test Time for each axis-(minutes)
10-20-10	2 ± 0.1 g level	3 ± 0.5	10 ± 0.1
20-50-20	0.036 ± 0.004 inch	3 ± 0.5	10 ± 0.1
50-500-50	5 ± 0.25 g level	3 ± 0.5	10 ± 0.1

3.9.6.2 Operated condition.

3.9.6.2.1 For 8857851-1 (Littlejohn). The switch shall withstand flight vibration through a frequency range of 75-1000-75 Hertz (Hz) at a peak of 10 ± 0.25 g followed by a vibration at 450 Hz at a peak of 10 ± 0.25 g without evidence of deterioration affecting functioning. During vibration, the electrical resistance between terminals 1-3 and 4-6 shall not exceed 0.030 ohms and the circuit continuity shall not be interrupted for more than 20 microseconds when tested as specified in 4.4.5.6.2.

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3.9.6.2.2 For 8857851-2 (Honest John). The switch shall withstand flight vibration through a frequency range of 75-1000-75 at a peak of 5 + 0.25 g followed by a vibration of 450 Hz at a peak of 5 ± 0.25 without evidence of deterioration affecting functioning. During vibration, the electrical resistance between terminals 1-3 and 4-6 shall not exceed 0.030 ohms and the circuit continuity shall not be interrupted for more than 20 microseconds when tested as specified in 4.4.5.6.2.

3.9.7 Storage.

3.9.7.1 High temperature. The switch shall withstand storage at 160 ± 5°F for 360 hours minimum. There shall be no evidence of damage such as swelling, cracking or loosening of parts or deterioration that affects functioning when tested as specified in 4.4.5.7.1.

3.9.7.2 Low temperature. The switch shall withstand storage at -40 ± 5°F for 360 hours min. There shall be no evidence of damage or deterioration that affects functioning when tested as specified in 4.4.5.7.2.

3.10 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460.

3.11 Workmanship.

3.11.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.11.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.11.3 Burrs. Parts shall be free from burrs.

3.11.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.11.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.11.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein.

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Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 50 complete assemblies (see dwg 8857851) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table IV. Tests shall be conducted in order specified in Table IV.

Table IV

First article inspection and tests

Inspection or Test	Number of Units 1-10	11-20	12-30	31-40	41-50	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)	X	X	X	X	X		X
2. Circuit resistance (safe) (see 3.5.1)	X	X	X	X	X		X
3. Insulation resistance (safe) (see 3.6.1)	X	X	X	X	X		X
4. Acceleration @80°F (safe) (see 3.9.1.1)	X	X	X	X	X		X
5. Acceleration @80°F (no load) (see 3.9.1.2)	X	X	X	X	X		X
6. Circuit resistance (operated) (see 3.5.2)	X	X	X	X	X		X
7. Insulation resistance (operated) (see 3.6.2)	X	X	X	X	X		X
8. Reset capability (see 3.8)	X	X	X	X	X		X
9. Sealing (see 3.9.4)	X	X	X	X	X		X
10. High temperature storage (see 3.9.7.1)		X	X				X
11. Acceleration @160°F (safe) (see 3.9.3.1)		X	X				X
12. Acceleration @160°F (operated) (see 3.9.3.2)		X	X				X
13. Low temperature storage (see 3.9.7.2)				X	X		X
14. Acceleration @-40°F (safe) (see 3.9.2.1)				X	X		X

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Table IV Continued

Inspection or Test	Number of Units 1-10	11-20	12-30	31-40	41-50	Dest	Non-Dest
15. Acceleration @-40°F (operated) (see 3.9.2.2)				X	X		X
16. Impact shock (safe) (see 3.9.5.1)		X	X			X	
17. Vibration (safe) (see 3.9.6.1)		X	X			X	
18. Acceleration @80°F (no load) (see 3.9.1.2)		X	X				X
19. Vibration (operated) (see 3.9.6.2)		X	X			X	
20. Current carrying capability (see 3.7)	X	X	X	X	X		X
21. Reset capability (see 3.8)	X	X	X	X	X		X
22. Current load operation (see 3.9.1.3)	X	X	X	X			X
23. Sealing (see 3.9.4)	X	X	X	X	X		X
24. Impact shock (operated) (see 3.9.5.2)					X	X	
25. Sealing (see 3.9.4)					X		X

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

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4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table V and Table VI), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table V
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table VI
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Shaft (see dwg 8839210).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Incorrect diameter (max)	Gage
	102. Incorrect flatness	Gage
	103. Incorrect surface finish	Gage
	104. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.11)	Visual

4.3.2.1.2 Spring, bias (see dwg 8857967).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Spring compression (see note 3)	Gage
	102. Objective evidence relating to material missing or improper (see 3.1)	Visual

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Minor:

201.	Incorrect load and spring rate	Gage
202.	Incorrect solid height	Gage
203.	Incorrect diameter	Gage
204.	Ends not squared	Visual
205.	Evidence of poor workmanship (see 3.11)	Visual

4.3.2.1.3 Weight setback (see dwg 8839206).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Mislocation of hole	Gage
102.	Incorrect diameter of hole (min)	Gage
103.	Mislocation of shaft groove	Gage
104.	Incorrect weight	Gage
105.	Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
201.	Incorrect concentricity counterbore with hole diameter	Gage
202.	Incorrect diameter of counterbore (min)	Gage
203.	Incorrect total length (max)	Gage
204.	Incorrect depth of counterbore (min)	Gage
205.	Incorrect depth of U shape groove (min)	Gage
206.	Improper or missing chamfer	Visual
207.	Evidence of poor workmanship (see 3.11)	Visual

4.3.2.1.4 Weight, timing (see dwg 8839205).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Mislocation of hole diameter	Gage
102.	Incorrect diameter of hole (min)	Gage
103.	Mislocation of shaft groove	Gage
104.	Mislocation of tapped hole (3)	Gage
105.	Incorrect pitch of gear teeth	Gage
106.	Incorrect length of gear rack	Gage
107.	Incorrect perpendicularity of gear teeth to hole diameter	Gage

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108. Incorrect weight.....Gage
109. Objective evidence relating to material missing or improper (see 3.1).....Visual

Minor:
201. Incorrect pitch diameter of tapped hole (3) (min).....Gage
202. Incorrect total length (max).....Gage
203. Incorrect pitch diameter of tapped hole (1) (min).....Gage
204. Incorrect depth of tapped hole (1) (Min).....Gage
205. Evidence of poor workmanship (see 3.11).....Visual

4.3.2.1.5 Sleeve (see dwg 8839234).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Incorrect inside diameter (min)	Gage
	102. Incorrect length (max)	Gage
	103. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.11)	Visual

4.3.2.1.6 Switch, firing actuator (left see dwg 8839212) (right see dwg 8839213).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Incorrect contour of profile	Gage
	102. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.11)	Visual

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4.3.2.1.7 Actuator, switch monitor (see dwg 8839211).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Incorrect contour of profile	Gage
	102. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.11)	Visual

4.3.2.1.8 Frame (see dwg 8839204).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Mislocation of insertion hole (4) for shaft	Gage
	102. Incorrect diameter of insertion hole (4) (min) for shaft	Gage
	103. Mislocation of mounting hole (2) for delay mechanism	Gage
	104. Mislocation of mounting hole (3) for actuator	Gage
	105. Mislocation of mounting hole (3) for switch	Gage
	106. Incorrect dia of guide hole for reset rod (2)	Gage
	107. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
	201. Incorrect pitch diameter of thread (min) (8)	Gage
	202. Incorrect inside length (min)	Gage
	203. Evidence of poor workmanship (see 3.11)	Visual

4.3.2.1.9 Delay assembly (modified) (see dwg 8839236).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	

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Major:
101. Inability of assembly to function (see note 3).....Visual
102. Lubrication missing from movable subassemblies as indicated on dwg 8839236.....Visual
103. Objective evidence relating to material missing or improper (see 3.1).....Visual

Minor:
201. Incorrect height of stake (2 places).....Gage
202. Evidence of poor workmanship (see 3.11).....Visual

4.3.2.1.10 Frame assembly (see dwg 8857968).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Components improper or missing	Visual/Manual
102.	Components damaged or loose	Visual
103.	Components improperly assembled	Visual
104.	Improper soldering (see 3.10)	Visual/Manual
105.	Missing or improper stake (4 places)	Visual
106.	Timing weight binds in escapement	Visual/Manual
107.	Pinion to rock clearance incorrect per note on dwg	Gage
108.	Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
201.	Evidence of poor workmanship	Visual

4.3.2.1.11 Acceleration switch assembly (see dwg 8857851).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Incorrect length of terminal (max)	Gage
102.	Terminals bent	Visual
103.	Incorrect total length (max)	Gage
104.	Incorrect diameter (max)	Gage
105.	Objective evidence relating to material missing or improper (see 3.1)	Visual

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Minor:
201. Marking improper, missing or illegible (see dwg 8857851).....Visual
202. Evidence of poor workmanship.....Gage/Visual

4.3.2.1.12 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Preservation, packing, and packaging not in accordance with MIL-STD-1169	Visual
	102. Marking not in accordance with MIL-STD-1169	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.11)	Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test Classification
a. Circuit resistance (safe)	Major
b. Insulation resistance (safe)	Major
c. Acceleration (safe)	Major
d. Acceleration (no load)	Major
e. Circuit resistance (operate)	Major
f. Insulation resistance (operate)	Major
g. Reset capability	Major
h. Sealing	Major
i. Impact shock (safe)	Major
j. Vibration (safe)	Major
k. Vibration (operate)	Major
l. Low temperature storage	Major

Of these tests, i, j and k are considered destructive and samples so tested shall not be returned to the lot. Desctuctive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Circuit resistance, insulation resistance, acceleration safe and no load, reset capability and sealing. Each switch in the lot shall be tested in accordance with 4.4.1, 4.4.2, 4.4.5.1.1, 4.4.5.1.2, 4.4.4 and 4.4.5.4. If any item fails to comply with 3.5.1, 3.5.2, 3.6.1, 3.6.2, 3.9.1.1, 3.9.1.2, 3.8 and 3.9.4, the item shall be rejected and removed from the lot.

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4.3.3.2 Impact shock (safe), Vibration (safe and operate) and low temperature storage. Four samples, one for each of these tests shall be selected at random from each lot of 50 switches or portion thereof produced. The samples shall be tested in accordance with 4.4.5.5.1, 4.4.5.6.1, 4.4.5.6.2 or 4.4.5.7.2. Following these tests each sample shall be seal tested in accordance with 4.4.5.4 and then subjected to three acceleration runs (test 2 of Table I or Table II, as applicable) in accordance with 4.4.5.1.2. If any sample fails to comply with 3.9.5.1, 3.9.6.1, 3.9.6.2, 3.9.7.2, 3.9.4 or 3.9.1.2 where applicable, the lot represented by the sample shall be rejected.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL). Els for assemblies are listed on the Equipment Tabulation, ET-8857851. When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Circuit resistance. This test shall be conducted by utilizing a low voltage measuring instrument and by applying a current of 100 ma max. The terminals of the switch shall be connected as specified in 3.5.

NOTE: CARE MUST BE EXERCISED TO AVIOD BREAKAGE OF THE GLASS SEALS.

4.4.2 Insulation resistance. This test shall be conducted by utilizing an instrument capable of measuring megohms and by applying a current not to exceed 5 ma. The terminals of the switch shall be connected as specified in 3.6.

4.4.3 Current carrying capability. The switch terminals 1 and 3 shall be connected to a device capable of producing the load current at the time cycles specified in 3.7 and Figure 2. The voltage drop between terminals 1 and 3 (or 4 and 6) shall be monitored during this test. The test shall be repeated using switch terminals 4 and 6.

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4.4.4 Reset capability. The switch shall be capable of being reset from the "operated condition" to the "safe condition" by pressing a soft rubber instrument against the flexible diaphram. Resetting can be detected by hearing the weights inside the switch release.
CAUTION: Do not apply pressure against the flexible diaphram with finger-nails or other hard objects.

4.4.5 Environmental testing (see 3.9).

4.4.5.1 Acceleration 80 ± 20°F

4.4.5.1.1 Safe condition. Acceleration tests shall be run on a centrifuge using a rotating, shockless fixture to bring the switch into proper orientation for operation. The major axis of the switch shall be placed perpendicular to the arm of the centrifuge within 2 degrees when the fixture is in its initial position. The fixture shall rotate the switch into the acceleration field after the proper acceleration is obtained. At this time the switch shall be oriented as shown on Figure 1. The major axis of the switch shall be parallel to the direction of acceleration within 2 degrees.

4.4.5.1.2 No load operation. Testing shall be conducted on a centrifuge using a rotating, shockless fixture to orient the switch for operation. The major axis of the device shall be perpendicular to the arm of the centrifuge within 2 degrees when the fixture is in its initial position. Upon attaining the specified acceleration the fixture shall be allowed to rotate the device so that its major axis is parallel to the direction of acceleration within 2 degrees. The fixture shall be capable of rotating the switch from initial position to final position in 0.047 to 0.150 seconds when the acceleration level is 15 g, and in 0.047 to 0.110 seconds when the acceleration level is 38 g. Switch operation time shall be measured from the time of release of the fixture from its initial position to opening of the (monitoring) contacts (circuit between terminals 2 and 5) and to closure of the activating contacts (circuits between terminals 1 and 3 and between 4 and 6). After the last centrifuge test in either Table I or Table II the switch shall be removed from the centrifuge in the operated condition and meet the requirements of 4.4.1 and 4.4.2.

4.4.5.1.3 Current load operation. This test shall be performed by using a centrifuge and a shockless test fixture of the type described in 4.4.5.1.2. The g level and load current to be applied are specified in 3.9.1.3. The resistance shall be monitored during the test.

4.4.5.2 Operating tests at -40°F.

4.4.5.2.1 Safe condition. Testing shall be conducted as specified in 4.4.5.1.1 except as follows. The switch shall be placed in an environmental test chamber, which has been stabilized at -40 ± 5°F per 6.4, for four (4) hours minimum and the temperature maintained at -40 ± 5°F.

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The switch shall then be rapidly transferred from the environmental test chamber to a temperature box mounted on the centrifuge. The centrifuge temperature box shall be stabilized at -40 ± 5°F per 6.4, and the switch shall remain in the temperature box for one hour, prior to testing, during that time the temperature shall be maintained at -40 ± 5°F.

4.4.5.2.2 No load operation. Testing shall be conducted as specified in 4.4.5.1.2 except as follows. The switch shall be placed in an environmental test chamber, which has been stabilized at -40 ± 5°F per 6.4 for four (4) hours minimum and the temperature maintained at -40 ± 5°F. The switch shall then be rapidly transferred from the environmental test chamber to a temperature box mounted on the centrifuge. The centrifuge temperature box shall be stabilized at -40 ± 5°F per 6.4, and the switch shall remain in the temperature box for one hour, prior to testing, during that time the temperature shall be maintained at -40 ± 5°F.

4.4.5.3 Operating tests at + 160°F.

4.4.5.3.1 Safe condition. Testing shall be conducted as specified in 4.4.5.1.1 except as follows. The switch shall be placed in a temperature test chamber, which has been stabilized at 160 ± 5°F per 6.4, for four (4) hours minimum and the temperature maintained at 160 ± 5°F. The switch shall then be rapidly transferred from the temperature test chamber to a temperature box mounted on the centrifuge. The centrifuge temperature box shall be stabilized at 160 ± 5°F, and the switch shall remain in the temperature box for one hour, prior to testing, during that time the temperature shall be maintained at 160 ± 5°F.

4.4.5.3.2 No load operation. Testing shall be conducted as specified in 4.4.5.1.2 except as follows: The switch shall be placed in a temperature test chamber, which has been stabilized at 160 ± 5°F per 6.4 for four (4) hours minimum and the temperature maintained at 160 ± 5°F. The switch shall then be rapidly transferred from the temperature test chamber to a temperature box mounted on the centrifuge. The centrifuge temperature box shall be stabilized at 160 ± 5°F, and the switch shall remain in the temperature box for one hour, prior to testing, during that time the temperature shall be maintained at 160 ± 5°F.

4.4.5.4 Sealing. The sealed switch shall be tested by utilizing a leak detection method involving the passage of tracer gas (helium) from one side of the presumed leak to the other side. The instrument shall provide a method of determining the leakage rate and the sensitivity of the instrument shall be capable of determining leakage rate specified in 3.9.4. If no leakage is detected during this test, immerse the switch at 80°F ± 20°F into water at 175 ± 15°F for 30 seconds minimum.

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Leaks will be evidenced by the growth of bubbles at any point. Small bubbles formed by the release of absorbed air shall not be interpreted as leaks. Bubbles indicating leakage shall be cause for rejection.

4.4.5.5 Impact shock.

4.4.5.5.1 Safe condition. The switch, firmly mounted to a suitable shock producing device shall be shocked as prescribed in 3.11.1. Shocks shall be applied at the rate at at least 1 shock per minute. The contacts shall be checked for continuity after this test. Following this test, the switch shall be tested as specified in 4.4.1, 4.4.2 and 4.4.5.4.

4.4.5.5.2 Operated condition. The switch shall be rigidly mounted to a suitable shock producing device and shocked in the direction opposite to that required for operation as specified in 3.9.5.2. Each shock shall be applied for the specified time. The contact resistance shall be monitored during this test with an instrument capable of detecting circuit openings of 20 microseconds minimum duration. Following this test, the switch shall be tested as specified in 4.4.1, 4.4.2 and 4.4.5.4.

4.4.5.6 Vibration.

4.4.5.6.1 Safe condition. The switch shall be rigidly mounted to a fixture capable of being readily positioned in each of three mutually perpendicular axes. This fixture shall be mounted on a suitable vibration producing apparatus. The switch shall be vibrated at the frequencies and amplitudes and for the time specified in Table III. The switch in the safe position, shall be tested as specified in 4.4.1, 4.4.2 and 4.4.5.4.

4.4.6.2 Operated condition. The switch shall be rigidly mounted to a fixture capable of being readily positioned in each of three mutually perpendicular axes. The fixture shall be mounted on a suitable vibration producing apparatus. The 75-1000-75 Hz frequency range shall be swept from the low value to high and back to low for a period of 3 $\pm$ 0.3 minutes followed by a 2 $\pm$ 0.2 minute vibration at 450 Hz in each axis. The contact resistance test shall be monitored and the instrument used shall be capable of detecting resistance not to exceed 0.030 ohm, utilizing a measuring current of 100 ma. (max). The circuit continuity (contact chatter) test shall be monitored and the instrument used shall be capable of detecting circuit openings of 20 microseconds minimum duration. The circuit resistance and the circuit continuity may be monitored during separate test runs. Following this test, the switch shall be tested as specified in 4.4.1, 4.4.2 and 4.4.5.4.

4.4.5.7 Storage.

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4.4.5.7.1 High temperature. The switch shall be placed in a temperature test chamber, stabilized at 160 ± 5°F per 6.4, fpr 360 hours minimum during which time the temperature shall be maintained at 160 ± 5°F. Following the 360 hours storage, the switch shall be transferred to a temperature chamber mounted on a centrifuge. The centrifuge temperature box shall be stabilized at 160 ± 5°F per 6.4, and the switch tested as specified in 4.4.5.3.1 and 4.4.5.3.2.

4.4.5.7.2 Low temperature. The switch shall be placed in a temperature test chamber, stabilized at -40 ± 5°F. per 6.4, for 360 hours minimum during which time the temperature shall be maintained at -40 ± 5°F. Following 360 hours storage, the switch shall be transferred to a temperature chamber mounted on a centrifuge. The centrifuge temperature box shall be stabilized at -40 ± 5°F per 6.4, and the switch tested as specified in 4.4.5.2.1 and 4.4.5.2.2.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The switch shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Level C. The switch shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.2 Packing.

5.2.1 Levels A and C. The switch shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-N-I.

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6.4 Temperature stabilization. The temperature is considered stabilized when the greatest peaks (both highest) and lowest) continually observed over 5 minutes min. do not exceed the nominal temperature plus tolerance specified.

6.5 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-2317 (Rev. 1) dated 15 February 1965 with Amendment 5 dated 16 January 1967.

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Preparing Activity:
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Project No.: 1135-A163

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